MILITARY SPECIFICATION

SPECTROMETER, FLUID ANALYSIS A/E35U-3

This amendment forms a part of Military Specification MIL-S-83129A, dated 22 September 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

Pages 11 and 12, paragraphs 3.9.1 and 3.9.1.1: Delete and substitute:

"3.9.1 Allowable deviation. The maximum allowable deviation for each element shall be:

$$Dmax \quad (Y \leq 100) = \sqrt{b^2 + (C\bar{X})^2}$$

$$Dmax \quad (Y > 100) = f\bar{X}$$

and the actual deviation of N (see 6.2) consecutive analyses of the same standard shall be calculated as follows:

$$Dact = \left[\frac{N \left(\Sigma Xi^2 \right) - \left(\Sigma Xi \right)^2}{N \ (N-1)} \right]^{1/2}$$

Where	Dmax	Maximum allowable deviation
	Dact	Actual deviation
	Y	Nominal value of the standard in ppm
	N	Number analyses per sample
	Xi	The determined contamination value of each analysis
	$\bar{X}$	The average determined contamination value of N consecutive analyses
	b,c,f	Element constants (see 3.9.1.1.1).

"3.9.1.1 Accuracy. The determination of contamination value of each element shall be made with the accuracy as expressed in the following equation:

$$\left| Y - \bar{X} \right| \max = \sqrt{.25 + (.08Y)^2} + \Delta Y$$

Where	Y	= The nominal value of the standard in ppm
	$\bar{X}$	= The average determined contamination value of N consecutive analyses
	N	= Number of analyses per sample (see 6.2)
	ΔY	= Accuracy tolerance of the standard oil in ±ppm (see 6.2)."

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AMENDMENT 1

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